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CSE471: System Analysis and Design
Assignment on Functional Requirements
Proposed Project Title: Smart Local Service Intelligence PLatform

Group No:07, CSE471 Lab Section:12, Spring26	
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Tech Stack:

- **Frontend:** React.js
- **Backend:** Node.js, Express.js
- **Database:** MongoDB
- **Authentication:** JWT (JSON Web Token)
- **Deployment:** Vercel (Frontend), Render / Railway (Backend)
- **Architecture:** MERN Stack

Functional Requirements:

Module 1 [Saud]

1. The system allows users to submit issues using natural language text input.
2. The system automatically categorizes issues into predefined categories such as Electrical, Plumbing, and AC-related problems.
3. The system provides a guided troubleshooting flow using a step-by-step decision-making process.
4. The system evaluates and displays a risk level indicator (Low / Medium / High) before suggesting any DIY action.
5. The system analyzes the submitted issue based on category, complexity, and risk level to determine whether it is suitable for DIY resolution.
6. The system generates an intelligent recommendation indicating whether the user should proceed with DIY troubleshooting or seek professional assistance.
7. The system calculates and displays a confidence score (%) representing the likelihood of successful DIY resolution.
8. The system maintains a record of all user-submitted issues, including problem description, issue category, suggested troubleshooting steps, risk level, and final outcome.
9. The system stores troubleshooting history to improve future recommendations and provide users access to past issue resolutions.

Module 2[Farhan]

1. The system maintains worker profiles including skill details, ratings, experience, and service area.
2. The system filters workers based on the user's location.
3. The system ranks workers using a rating-based ranking algorithm.
4. The system allows workers to update their real-time availability status.
5. The system allows users to send service requests to workers.
6. The system supports an emergency priority request option for urgent service needs.

Module 3[Deema]

1. The system provides an estimated service cost before booking.
2. The system displays a transparent cost breakdown including labor, parts, and urgency charges.
3. The system allows users to rate and review workers after service completion.
4. The system tracks service warranty duration for completed jobs.
5. The system allows users to submit disputes regarding service quality or pricing.
6. The system provides a workflow for dispute resolution managed by the admin.